# Lab 3: Linear Regression — Model Specification, Comparison, and Interpretation

## Overview

In this lab, you will apply linear regression to investigate the factors associated with employee ratings of their workplace. Rather than estimating a single regression model, you will develop and compare a series of competing models, evaluate their performance, examine regression diagnostics, and interpret the results of your preferred model.

Your goal is to produce a **reproducible statistical report** that demonstrates not only how to estimate regression models, but also how to make and justify decisions about model specification and selection.

You will complete the analysis in **R**, document your workflow in **Quarto**, and publish the rendered report using **GitHub Pages**.

### Learning objectives

By the end of this lab, you should be able to:

1. Specify multiple linear regression models based on substantive predictions.
2. Estimate and compare nested regression models using F-tests.
3. Extract model-level statistics using `broom::glance()`.
4. Compare models using $R^2$, adjusted $R^2$, $\Delta R^2$, and RMSE.
5. Create a combined model-comparison table using `gt`.
6. Evaluate regression assumptions and identify potentially influential observations.
7. Select a preferred model and justify that decision.
8. Extract, report, and interpret regression coefficients using `broom::tidy()`.
9. Communicate your analysis in a reproducible Quarto report.

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# Part 1: Configure your project and examine the data

### 1.1. Create your project

Create a new GitHub repository for this lab, following the workflow established in Lab 1.

Your repository should contain:

- An RStudio project connected to your GitHub repository.
- An `index.qmd` file containing your analysis and written report.
- A `_quarto.yml` file configured to render your report to the `docs/` directory.

- A `docs/` directory containing the rendered website.

Your analysis must be reproducible from the files in your repository.

**Important:** As in Lab 1, you should distinguish between commands used for *interactive exploration* in the R Console and commands required to *reproduce your results* in Quarto.

You may use the Console to inspect datasets, experiment with model specifications, and troubleshoot errors. However, every command necessary to generate the results presented in your report must appear in your Quarto document.

## 1.2. Load the required packages

You will need the following packages:

- `dplyr` and `tidyr` for data manipulation.
- `ggplot2` for visualization.
- `broom` for extracting regression results.
- `performance` for regression diagnostics and model comparisons.
- `gt` for formatting tables.

Install any missing packages in the **R Console**. Load the packages in a setup chunk near the beginning of your Quarto document.

## 1.3. Load the dataset

We will use the built-in R dataset `attitude`, which contains data from **30 departments**.

The outcome variable is `rating`, an overall workplace rating. Several other variables summarize employee attitudes toward different aspects of the workplace.

For this lab, we will focus on four variables:

| Variable | Description | Role |
|---|---|---|
| `rating` | Overall workplace rating | Outcome |
| `complaints` | Handling of employee complaints | Predictor |
| `learning` | Opportunity to learn | Predictor |
| `raises` | Pay increases | Predictor |

The scores are department-level summaries of employee responses. Consequently, the observations represent **departments, not individual employees**.

Load the data and create an analysis dataset containing these four variables.

### 1.4. Explore the data

Before estimating any models, familiarize yourself with the variables and their distributions.

**In the R Console:**

- Inspect the first several observations.
- Examine the dataset's structure.
- Calculate descriptive statistics.
- Explore the distributions of the variables.
- Examine the relationships between `rating` and each predictor.

**In your Quarto report:**

Include:

1. A formatted table reporting the mean, standard deviation, minimum, and maximum for each analysis variable.
2. A histogram of `rating`, with an overlaid density curve.
3. At least two scatterplots showing relationships between `rating` and predictors, with fitted regression lines.
4. A short paragraph describing the most important patterns you observe.

Your discussion should consider whether the relationships appear approximately linear, whether there are unusual observations, and whether the exploratory results are consistent with your expectations.

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## Part 2: Specify competing regression models

### 2.1. Develop your predictions

Before fitting your models, write a short paragraph explaining why you expect the three predictors to be associated with overall workplace ratings.

Consider the following questions:

- Why might the handling of employee complaints be associated with overall workplace ratings?
- Could opportunities for learning explain additional variation beyond complaint handling?
- Might pay increases explain further variation after accounting for complaints and learning?

These predictions should motivate your model specifications.

### 2.2. Specify four nested models

Estimate the following models:

| Model | Predictors | Purpose |
|---|---|---|
| M0: Null | Intercept only | Establish a baseline |
| M1: Complaints | `complaints` | Evaluate complaint handling |
| M2: Complaints + Learning | `complaints`, `learning` | Evaluate the additional contribution of learning |
| M3: Expanded | `complaints`, `learning`, `raises` | Evaluate the additional contribution of pay increases |

These models are **nested**, meaning that each successive model includes all predictors from the preceding model.

Use `lm()` to estimate each model, assigning each fitted model to a clearly named object.

### 2.3. Explore your fitted models

In the **R Console**, use `summary()` to inspect the results of each fitted model.

Pay particular attention to:

- The estimated regression coefficients.
- The direction and magnitude of predictor associations.
- The model $R^2$ and adjusted $R^2$.
- How the coefficients change as additional predictors are included.

In your Quarto report, briefly explain how the interpretation of the `complaints` coefficient changes between M1, M2, and M3.

You do not need to reproduce the complete `summary()` output for every model.

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## Part 3: Compare model performance

This is the central analytical component of the lab.

You will compare the four models using **nested-model F-tests** and several measures of overall model performance.

## 3.1. Conduct nested-model F-tests

Use `anova()` to compare all four models in a single call, in the order M0, M1, M2, M3.

The resulting table will allow you to evaluate three questions:

1. Does complaint handling explain significantly more variation than the intercept-only model?
2. Does adding learning opportunities improve model fit beyond complaint handling?
3. Does adding pay increases improve model fit beyond complaints and learning?

Retain the following information from the ANOVA results:

- Residual degrees of freedom.
- Change in degrees of freedom.
- F-statistic for the nested-model comparison.
- p-value for the nested-model comparison.

Use `select()` and `rename()` to organize these results into a clearly labelled data frame.

## 3.2. Extract model-level performance statistics

Use `glance()` from the `broom` package to extract model-level statistics for each of the four fitted models.

Combine the results into one data frame using `rbind()` or `bind_rows()`.

Include a column identifying each model.

For this lab, focus on:

- **$R^2$:** The proportion of variation in overall workplace ratings explained by the model.
- **Adjusted $R^2$:** A measure of explained variation that accounts for the number of predictors.
- **RMSE:** The root mean squared error, expressed in units of the outcome.

Calculate RMSE directly from the residuals of each fitted model.

You may encounter AIC and BIC in the output from `glance()` or other model-comparison functions. You are **not required to interpret these metrics** in this lab.

## 3.3. Calculate changes in explained variance

Calculate the change in $R^2$ between successive models:

- M1 − M0.
- M2 − M1.

- M3 − M2.

Use `mutate()` and `lag()` to add a $\Delta R^2$ column to your model-performance data frame.

The null model should have a missing $\Delta R^2$ value because there is no preceding model.

## 3.4. Create a combined model-comparison table

Combine the model-performance statistics with the nested-model test results.

You may use `cbind()` because both sets of results contain the same four models in the same order.

Use `gt()` to create a professionally formatted table containing:

| Column | Meaning |
|---|---|
| Model | Model identifier |
| $R^2$ | Overall explained variance |
| Adjusted $R^2$ | Explained variance adjusted for complexity |
| RMSE | Root mean squared error |
| $\Delta R^2$ | Additional explained variance relative to the preceding model |
| Residual df | Residual degrees of freedom |
| Nested df | Degrees of freedom added by the new predictor(s) |
| Nested F | F-statistic comparing the model with its predecessor |
| Nested p | p-value for the nested comparison |

Format the numerical results to an appropriate number of decimal places.

Your table should clearly distinguish **overall model performance** from **incremental improvements in fit**.

## 3.5. Interpret the comparisons

Write a paragraph addressing the following:

1. Which model explains the largest proportion of variation in `rating`?
2. How does adjusted $R^2$ change as predictors are added?
3. How much additional variation does each successive model explain?
4. Which additions produce statistically significant improvements in fit?
5. Do the reductions in RMSE appear substantively meaningful?

Do not simply identify the model with the largest $R^2$. Explain what the pattern of results suggests about the contribution of each predictor.

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# Part 4: Evaluate regression diagnostics

## 4.1. Examine the assumptions

Select the most promising model based on your comparisons in Part 3.

Use R's standard regression diagnostic plots and/or `performance::check_model()` to examine its assumptions.

Evaluate:

- **Linearity:** Is the relationship between predictors and the outcome adequately represented by the model?
- **Normality:** Are the residuals approximately normally distributed?
- **Homoscedasticity:** Is residual variability reasonably consistent across fitted values?
- **Influential observations:** Are there observations that may disproportionately influence the fitted model?
- **Multicollinearity:** Are predictors so strongly related that their unique contributions may be difficult to estimate precisely?

You may also use `check_collinearity()` to investigate multicollinearity.

## 4.2. Report your diagnostic findings

Include at least one set of diagnostic plots in your Quarto report.

Write a short paragraph explaining:

1. Which assumptions appear reasonably satisfied.
2. Whether any potential problems are evident.
3. Whether those problems affect your confidence in the model.

Avoid treating regression assumptions as a mechanical checklist. Explain what the diagnostic results imply for the interpretation of your findings.

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# Part 5: Select and report your preferred model

## 5.1. Select a model

Identify the model you believe provides the most defensible representation of the relationship between the workplace attitude variables and overall ratings.

Your decision should consider:

- Nested-model F-tests.
- $\Delta R^2$.
- Adjusted $R^2$.
- RMSE.
- Model complexity.
- Diagnostic findings.
- Substantive interpretability.

You are **not required to select the most complex model**.

Write a paragraph explaining why your preferred model is more appropriate than the alternatives.

## 5.2. Create a coefficient table

Use `tidy()` from `broom` to extract the regression coefficients from your selected model.

Request 95% confidence intervals.

Use `gt()` to create a formatted coefficient table containing:

- Predictor name.
- Unstandardized coefficient estimate.
- Standard error.
- 95% confidence interval.
- p-value.

Do not simply paste the output from `summary()` into your report.

## 5.3. Interpret the coefficients

Write a paragraph interpreting the regression coefficients in your selected model.

For each included predictor:

1. Describe the direction of the association.
2. Explain the coefficient in the units of the original variables.
3. Explain what is being held constant, if the model includes multiple predictors.
4. Consider the uncertainty reflected in the confidence interval.

Use **inline R code referencing your `tidy()` results** so that the numerical values in your written interpretation update automatically when the model is re-estimated.

For example, you can use `filter()` and `pull()` to retrieve the estimate and confidence interval limits for a particular predictor.

You should not manually type coefficient estimates or confidence intervals into your narrative.

### 5.4. Summarize model performance

Report your selected model's:

- $R^2$.
- Adjusted $R^2$.
- RMSE.

Explain these statistics in the context of overall workplace ratings.

Use inline R code where appropriate to keep the narrative synchronized with the model results.

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## Part 6: Discuss your findings and limitations

What do your results suggest about the relative importance of complaint handling, learning opportunities, and pay increases for overall workplace ratings?

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## Part 7: Render, publish, and submit

### 7.1. Verify reproducibility

Before submitting your work:

1. Save your Quarto document.
2. Restart your R session.
3. Render your Quarto document from a clean session.
4. Confirm that all tables, figures, and inline statistical results are generated successfully.
5. Inspect the rendered HTML report for formatting problems.

Your report should not depend on objects created manually in the R Console.

### 7.2. Publish to GitHub Pages

Following the workflow from Lab 1:

1. Commit your completed Quarto document.
2. Commit the rendered output in `docs/`.
3. Push your changes to GitHub.
4. Verify that your GitHub Pages website displays the completed report.

### 7.3. Submit your work

Submit the following links:

- **GitHub repository:** Your source code and project files.
- **GitHub Pages website:** Your rendered statistical report.

Your report should contain all required analyses, tables, figures, interpretations, and conclusions.

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# Final submission checklist

## Data exploration

- Dataset loaded and analysis variables identified.
- Descriptive statistics presented in a formatted table.
- Outcome distribution visualized.
- At least two predictor–outcome relationships visualized.
- Exploratory patterns discussed.

## Model specification and comparison

- Four prescribed regression models estimated.
- Nested-model F-tests conducted.
- Model-level statistics extracted using `glance()`.
- RMSE calculated for each model.
- $\Delta R^2$ calculated between successive models.
- Combined model-comparison table created using `gt()`.
- Model-comparison results interpreted.

## Diagnostics and model selection

- Regression diagnostics examined.
- Diagnostic plots included.
- Potential assumption violations discussed.
- Preferred model selected and justified.

## Reporting and interpretation

- Coefficient table generated using `tidy()` and `gt()`.
- Coefficients interpreted in context.

- $R^2$, adjusted $R^2$, and RMSE reported.
- Numerical interpretations generated using inline R code.
- Substantive conclusions and limitations discussed.

## Reproducibility and submission

- Quarto document renders from a clean R session.
- Code, tables, plots, and written interpretations are synchronized.
- Changes committed and pushed to GitHub.
- GitHub Pages website verified.
- Repository and website links submitted.

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